Buffering, alkalinity and silica

- **Acidity** of a solution is its *capacity to donate protons*. It can be determined by titrating the solution with a strong base (e.g. NaOH).
- **Alkalinity** of a solution is its *capacity to accept protons*. It can be determined by titrating the solution with a strong acid (e.g. HCl).
- Acidity is an important property because high acidity enables minerals and metals to be dissolved and is often accompanied by high ***total dissolved solids*** (TDS) which means the water is unsuitable for drinking.
- Strong Acids more readily release H^+ into water, they more fully dissociate
- Strong Bases more readily release OH^- into water, they more fully dissociate
- Remember that strength DOES NOT EQUAL concentration.

Acid and Base dissociation

- For any acid, it's reaction in water is given as:

$$HA + H_2O \leftrightharpoons H^+ + A^- + H_2O$$

- The equilibrium expression, K (often denotes K_a or K_b for acids or bases)

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

- Strength of an acid or base is related to the dissociation constant, the higher the K_a or K_b strong acid/base.
- Remember from Lecture 1: $pK_a = -\log K_a$, so lower pK_a = stronger acid (or base).
- Acids with more than one proton dissociate stepwise with more than one pK_a

e.g.

$H_3PO_4 \leftrightharpoons H^+ + H_2PO_4^-$ pK_1

$H_2PO_4^- \leftrightharpoons H^+ + HPO_4^{2-}$ pK_2

$HPO_4^{2-} \leftrightharpoons H+ + PO_4^{3-}$ pK_3

- Same stepwise dissociation for bases – see Table in Faure for K_a and K_b values

pH scale

- Water will dissociate into ions: $H_2O \leftrightharpoons OH^- + H^+$

$$K = \frac{[H^+][OH^-]}{[H_2O]} = 1 \times 10^{-14} \text{ at } 25^\circ C \text{ and } 1 \text{ atm}$$

$$pK = -\log(1 \times 10^{-14}) = 14$$

- The convention is that $[H_2O] = 1.0$ so $[H^+] = [OH^-] = 1 \times 10^{-7}$ mol/L
- The activity of hydrogen ions is commonly expressed as the pH defined as:

pH = -log $[H^+]$

- So pH of pure water at 25°C is 7.0 (NB changes with T at 0°C pure water has a pH of 7.47).

When $[H^+] < 10^{-7}$ mol/L then pH >7 ***basic***

When $[H^+] > 10^{-7}$ mol/L then pH <7 ***acidic***

- Also : **pOH = -log$[OH^-]$**

pK = pOH + pH = 14

- Most natural waters have a pH range between 5-9

pH buffers

- Addition of an acid or a base to a solution should change the pH.
- Some solutions resist pH changes because they can consume H^+ or OH^- ions being added by reactions with dissolved species in the solution.
- A solution which is able to maintain its pH when an acid or base is added is called a **pH buffer.**
- The **buffering capacity** of a solution is a measure of how many H^+ or OH^- ions it can consume without a significant change in pH.
- A common pH buffer in the natural environment is **seawater** because it contains a significant amount of HCO_3^- and $CaCO_3$. The buffering capacity of these two species maintains the pH of seawater at about 7.9 (pre-industrial CO_2 levels it was 8.2).
- The oceans contain essentially an infinite amount of $CaCO_3$ particles suspended in surface waters and in sediments and any excess (or increase in) H^+ can be buffered by dissolution of $CaCO_3$

$$CaCO_3(s) + H^+ \leftrightharpoons Ca^{2+} + HCO_3^-$$

- Ocean acidification is reducing the calcification rates of shell building marine organisms (effectively working against the gradient established in the above chemical reaction).

Alkalinity

- **Alkalinity** is the capacity of a solution to neutralize or “buffer” acid.
- In most natural waters carbonate [CO_3^{2-}] and bicarbonate [HCO_3^-] anions are most common, so ***total alkalinity*** is approximately equal to the ***carbonate alkalinity***.
- The alkalinity (and acidity) for many natural waters can be expressed in terms of carbonic acid species and strong acids and bases (which dissociate completely).

$$C_{alk} = [HCO_3^-] + 2[CO_3^{2-}] + [OH^-] - [H^+]$$

- Alkalinity can increase the pH (more basic). When $CaCO_3$ dissolves in water, the carbonate can react with water to form bicarbonate, which produces hydroxide:

$$CaCO_3\ (s) \leftrightharpoons Ca^{2+} + CO_3^{2-}$$

$$CO_3^{2-} + H_2O \leftrightharpoons HCO_3^- + OH^-$$

- OH^- is a strong base, so increase in OH^- concentration will cause the pH to rise.

Henderson-Hasselbalch Equation

$$pH = pK + \log \frac{[A^-]}{[HA]}$$

- When acid or base is added to buffered system with a pH near pK the pH will not change much.
- When the pH is further from the pK, additions of acid or base will change the pH much more.
- Dissolution of albite feldspar ($NaAlSi_3O_8$) is a base-generating reaction:

$$NaAlSi_3O_8 + 2H_2O = 4OH^- + Na^+ + Al^{3+} + 3SiO_{2(aq)}$$

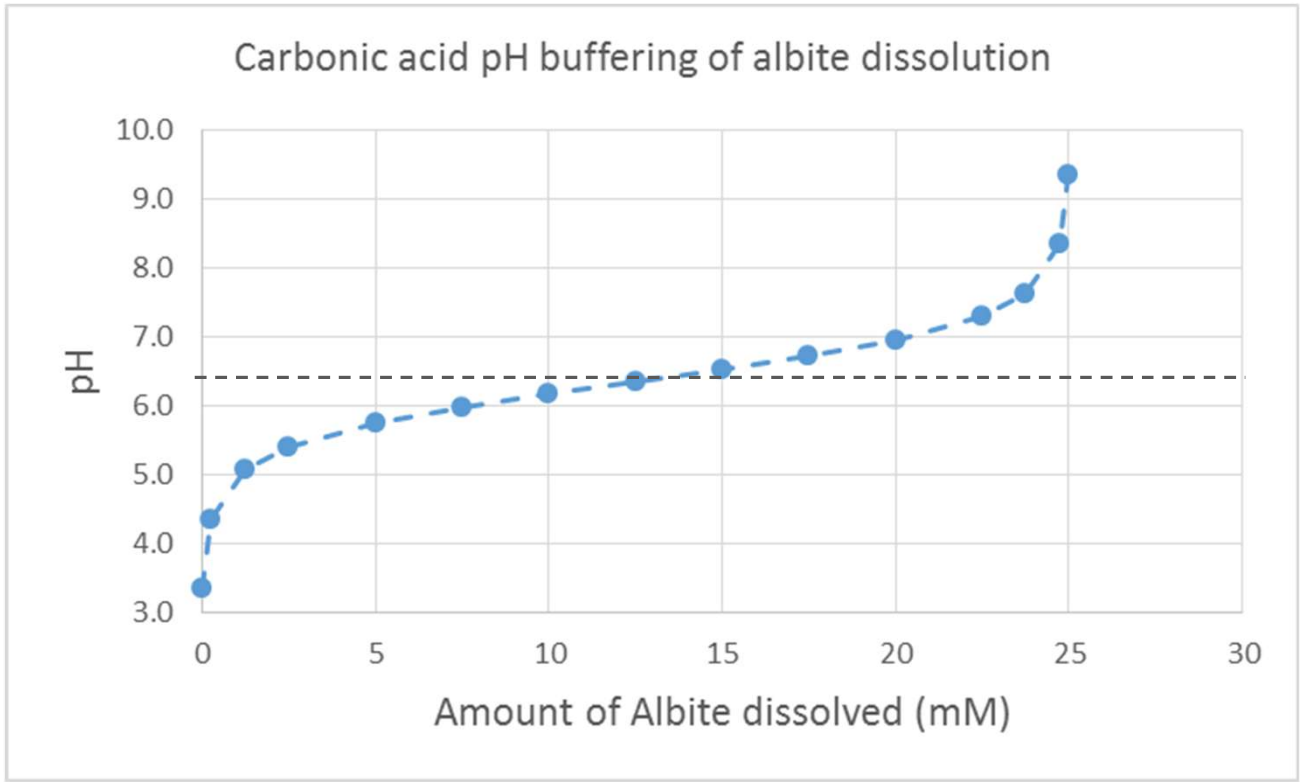
- Think of albite dissolution as titrating OH^- into solution – e.g. dissolve 20 millimoles albite = 80 millimoles OH^-
  - We know that in the absence of a buffer (low acidity) 80millimoles = 0.08 mol OH^- = -log(0.08) = pOH = 1.1
  - Then pH = pK – pOH = 14 – 1.1 =<u>12.9</u>
- What happens to the pH of a groundwater containing 100 millimoles (mM) H_2CO_3 ?
  - pK_1 for H_2CO_3 = 6.35

Buffering the pH change

$$pH = pKa + \log \frac{[HCO_3^-]}{[H_2CO_3]}$$

- 100 mM $H_2CO_{3(aq)}$, as the pH increases, some of this starts turning into HCO_3^-
- After 12.5 mM albite react to form 50 mM OH^-:
  - pH = 6.35 + log(50/50) = 6.35 + 0 = 6.35

  After 20 mM albite react to form 80 mM OH^-:
  - pH = 6.35 + log(80/20) = 6.35 + 0.6 = 6.95



Maximum buffering ±1pH unit from pK (dissociation constant)

Importance of alkalinity

- Low alkalinity of a water can result in rapid changes to pH, if an acid enters the water.
- Alkalinity is important for aquatic organisms because it protects them against rapid/transient changes in pH ("acid shock").
- Alkalinity is important for two of the most severe environmental impacts of industrialization linked to rising CO_2 in the atmosphere – **acid rain** and **ocean acidification**.

- Carbon dioxide and water are converted to carbonic acid, this can then dissociate to provide bicarbonate and carbonate:

$$CO_2 + H_2O \leftrightharpoons H_2CO_3 \leftrightharpoons H^+ + HCO_3^- \leftrightharpoons 2H^+ + CO_3^{2-}$$

- Notice in the reaction that as carbonic acid (H_2CO_3) reacts to form carbonate (CO_3^{2-}), **two** hydrogen ions H^+ are released into the water.

- Conversion of carbon dioxide to carbonic acid causes a **decrease in pH (increase in H^+).**

Solubility of silica

- Silicon dioxide (SiO_2) is a solid that can be amorphous or have a variety of crystalline forms including quartz (a very abundant mineral in the Earth's crust), cristobalite or tridymite.

- Amorphous silicon dioxide (silica) reacts with water to form silicic acid

- Most silicic acid forms during weathering of common rock-forming aluminosilicates such as feldspars and micas. Very little is formed by the dissolution of quartz itself.

- When its solubility is exceeded it forms amorphous silica which precipitates out as a gelatinous material. The water is slowly expelled and the silica crystallises as ***chert.***

Stepwise dissolution of amorphous silica

- The following reaction describes the dissolution of amorphous silica:

$$SiO_2 + 2H_2O \leftrightharpoons H_4SiO_4 \qquad K = 1.82 \times 10^{-3}$$

- Each mole of SiO_2 produces one mole of H_4SiO_4 so at equilibrium in neutral conditions with SiO_2 = 60.084 amu: grams of SiO_2 = 60.084 × (1.82×10^{-3}) = 0.109 g/Kg = 109 ppm

- This concentration does not take into consideration the stepwise dissociation of silicic acid:

$$H_4SiO_4 \leftrightharpoons H_3SiO_4^- + H^+ \qquad K_1 = 1.95 \times 10^{-10}$$
$$H_3SiO_4^- \leftrightharpoons H_2SiO_4^{2-} + H^+ \qquad K_2 = 5.25 \times 10^{-14}$$
$$H_2SiO_4^{2-} \leftrightharpoons HSiO_4^{3-} + H^+ \qquad K_3 = 1.38 \times 10^{-10}$$
$$HSiO_4^{3-} \leftrightharpoons SiO_4^{4-} \qquad K_4 = 7.94 \times 10^{-14}$$

- Thus, the concentration of silica in solution needs to include all of the Si-bearing ions in solution At neutral conditions, pH = 7, the dissociation constant for silicic acid is given by:

$$K_1 = \frac{\left[H^+\right]\left[H_3SiO_4^-\right]}{\left[H_4SiO_4\right]} = 1.95 \times 10^{-10} \;;\; \left[H^+\right] = 1 \times 10^{-7};$$

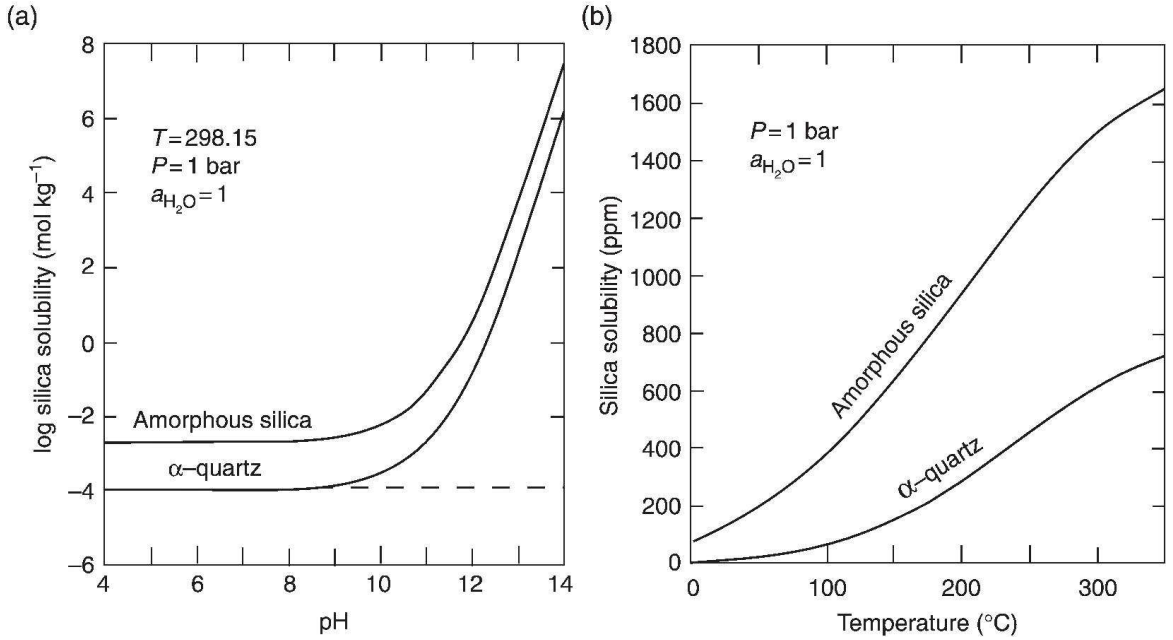
$$\left[H_3SiO_4^-\right] = \frac{(1.95 \times 10^{-10})(1.82 \times 10^{-3})}{1 \times 10^{-7}} = 3.55 \times 10^{-6}$$

- This is 500 times lower than the activity of silicic acid, so none of the other Si-bearing ions contribute to the solubility of amorphous silica.

pH and silica solubility

- At pH = 8.0 the activity of $H_4SiO_4^-$ rises by an order of magnitude because $[H^+]$ decreases by factor of ten from 10^{-7} to 10^{-8}.

- Up to pH = 9 (the maximum pH in most natural aqueous environments) the solubility is almost entirely due to K_1, at higher pH values, solubility increases exponentially as successive K terms become important.



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- The solubility of amorphous silica is also highly dependent on temperature (see Misra for equations).

Precipitation of silica from solution

- A relatively small decrease in pH of a basic solution containing silicic acid may reduce its solubility to form a supersaturated solution from which amorphous silica will precipitate.

- e.g. a decrease in pH from 8.5 to 8.4 can result in precipitation of 1.37 mg L^{-1} of silica. This amount magnifies rapidly in the natural environment because we are often dealing with relatively large volumes of aqueous fluids (groundwaters) and high flow rates.

- Quartz is a very common mineral in the Earth's crust but it is not an important source of silicic acid because it is highly insoluble (about 20x lower solubility than amorphous silica)

- Quartz solubility = 0.0059 g L^{-1} = 5.9 ppm.

- For amorphous silica the corresponding value is 109 ppm

Silica in natural waters

Dissolved constituent	Concentration: Average river water[a] (ppm)	Concentration: Average seawater[b] (ppm)
Cl^-	7.8	19,000
Na^+	6.3	10,500
Mg^{2+}	4.1	1300
SO_4^{2-}	11.2	2650
Ca^{2+}	15.0	400
K^+	2.3	380
HCO_3^-	58.4	140
Br^-	–	65
CO_3^{2-}	–	18
Sr^{2+}	–	8
SiO_2	13.1	6
Organic C	9.6	0.5

[a]Livingstone (1963); [b]Goldberg (1957).
[c]Total mass of ocean water used in the calculation = 1.37×10^{21} kg.
[d]Total river discharge into the oceans per year used in the calculatio

Misra: Table 13.7

River water is **supersaturated** with respect to quartz but **undersaturated** in amorphous silica. Quartz does not precipitate directly from aqueous solutions because the rate of reaction is very slow (kinetics).

Seawater has a lower dissolved silica content than river water (even though it is more alkaline, conditions under which silica is more soluble)

Why?

Equilibria in the system K-Al-Si-H_2O

$$2KAlSi_3O_8 \text{ (microcline)} + 2H^+ + 9H_2O \leftrightharpoons 2K^+ + 4H_4SiO_4 + Al_2Si_2O_5(OH)_4 \text{(kaolinite)}$$

$$K = \frac{[H_4SiO_4]^4[K^+]^2}{[H^+]^2} \Rightarrow \log K = 4\log[H_4SiO_4] + 2\log[K^+] - 2\log[H^+]$$

$$\Rightarrow \frac{\log K}{2} = 2\log[H_4SiO_4] + \log\frac{[K^+]}{[H^+]}$$

$$\log\frac{[K^+]}{[H^+]} = -2\log[H_4SiO_4] + \frac{\log K}{2}$$

$$4\log(a) + 2\log(b) - 2\log(c)$$

$$\log\left(\frac{a^4b^2}{c^2}\right) \rightarrow \log\left(\frac{a^2b}{c}\right) \rightarrow 2\log(a) + \log(b) - \log(c)$$

$$\rightarrow 2\log(a) + \log\left(\frac{b}{c}\right)$$

Can now plot $\log[H_4SiO_4]$ on the "x" axis and $\log([K^+]/[H^+])$ on the "y" axis. Then the equation above has the form of a straight line y = mx + c with a *slope m of -2 and a "y" intercept of log K/2.*

Table 14.1 Stability Relations Among Microcline, Kaolinite, Gibbsite, and Amorphous Silica in the Presence of Water at 25 °C and 1 atm Pressure[a]

Microcline–kaolinite

$$2\ KAlSi_3O_8 + 9\ H_2O + 2\ H^+ \rightleftharpoons Al_2Si_2O_5(OH)_4 + 2\ K^+ + 4\ H_4SiO_4$$

$$\Delta G_R^\circ = +7.103\ \text{kcal} \qquad K = 10^{-5.21}$$

$$\log \frac{[K^+]}{[H^+]} = -2 \log [H_4SiO_4] - 2.60 \tag{1}$$

Kaolinite–gibbsite

$$Al_2Si_2O_5(OH)_4 + 5\ H_2O \rightleftharpoons 2\ Al(OH)_3 + 2\ H_4SiO_4$$

$$\Delta G_R^\circ = +12.755\ \text{kcal} \qquad K = 10^{-9.35}$$

$$\log [H_4SiO_4] = -4.68 \tag{2}$$

Microcline–muscovite

$$3\ KAlSi_3O_8 + 12\ H_2O + 2\ H^+ \rightleftharpoons KAl_3Si_3O_{10}(OH)_2 + 2\ K^+ + 6\ H_4SiO_4$$

$$\Delta G_R^\circ = +16.184\ \text{kcal} \qquad K = 10^{-11.865}$$

$$\log \frac{[K^+]}{[H^+]} = -3 \log [H_4SiO_4] - 5.93 \tag{3}$$

Muscovite–kaolinite

$$2\ KAl_3Si_3O_{10}(OH)_2 + 3\ H_2O + 2\ H^+ \rightleftharpoons 3\ Al_2Si_2O_5(OH)_4 + 2\ K^+$$

$$\Delta G_R^\circ = -11.059\ \text{kcal} \qquad K = 10^{+8.11}$$

$$\log \frac{[K^+]}{[H^+]} = +4.05 \tag{4}$$

Muscovite–gibbsite

$$KAl_3Si_3O_{10}(OH)_2 + 9\ H_2O + H^+ \rightleftharpoons 3\ Al(OH)_3 + K^+ + 3\ H_4SiO_4$$

$$\Delta G_R^\circ = +13.603\ \text{kcal} \qquad K = 10^{-9.97}$$

$$\log \frac{[K^+]}{[H^+]} = -3 \log [H_4SiO_4] - 9.97 \tag{5}$$

Solubility limit of amorphous silica

$$SiO_2(\text{amorph.}) + 2\ H_2O \rightleftharpoons H_4SiO_4$$

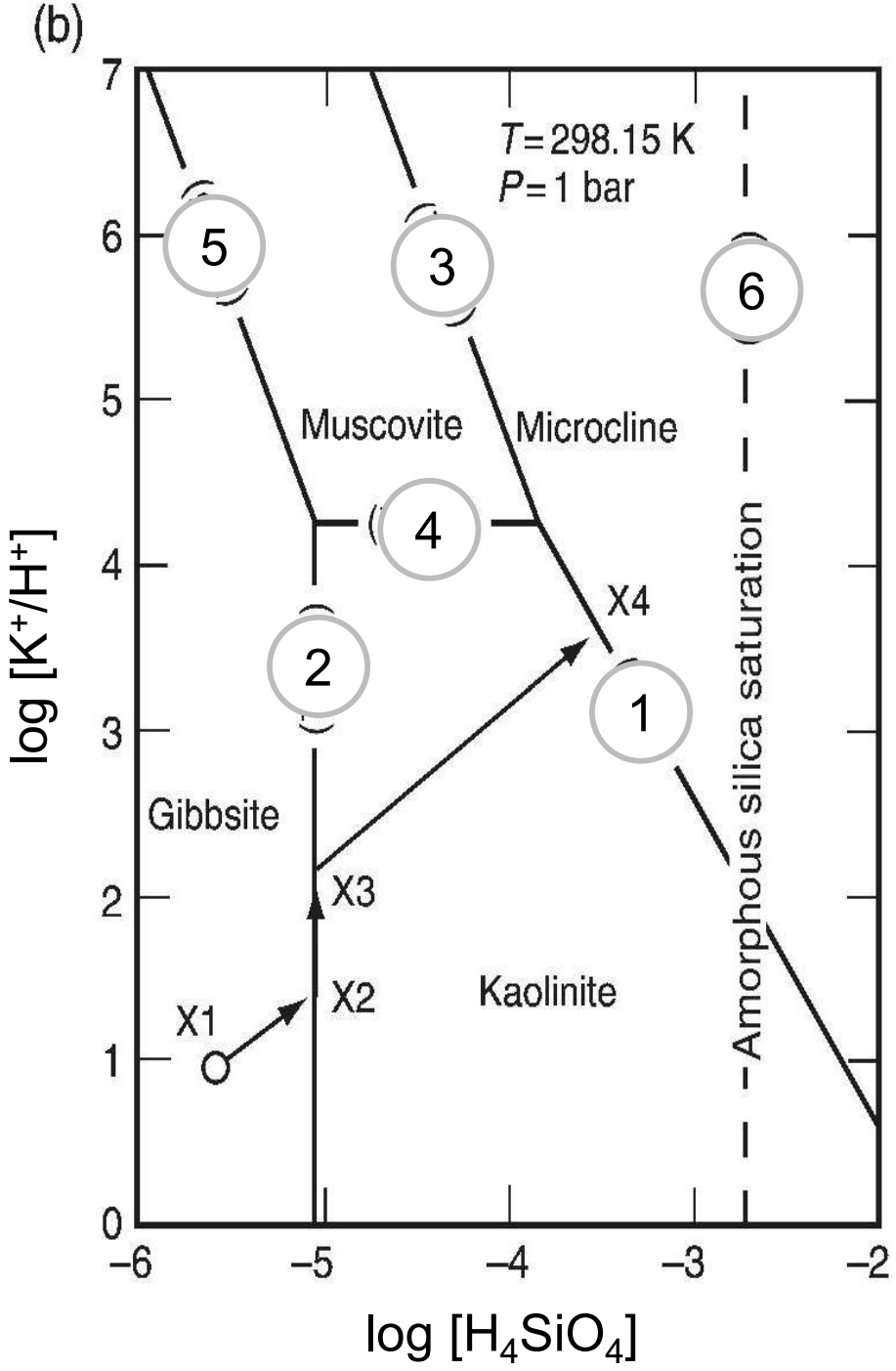
$$\Delta G_R^\circ = +3.604\ \text{kcal} \qquad K = 10^{-2.64}$$

$$\log [H_4SiO_4] = -2.64 \tag{6}$$

In order to complete the system, reactions need to be considered between: K-feldspar, kaolinite, muscovite, gibbsite and amorphous silica

All reactions 1-4 consume H^+ and liberate K^+ and silicic acid.

Mineral stability diagram



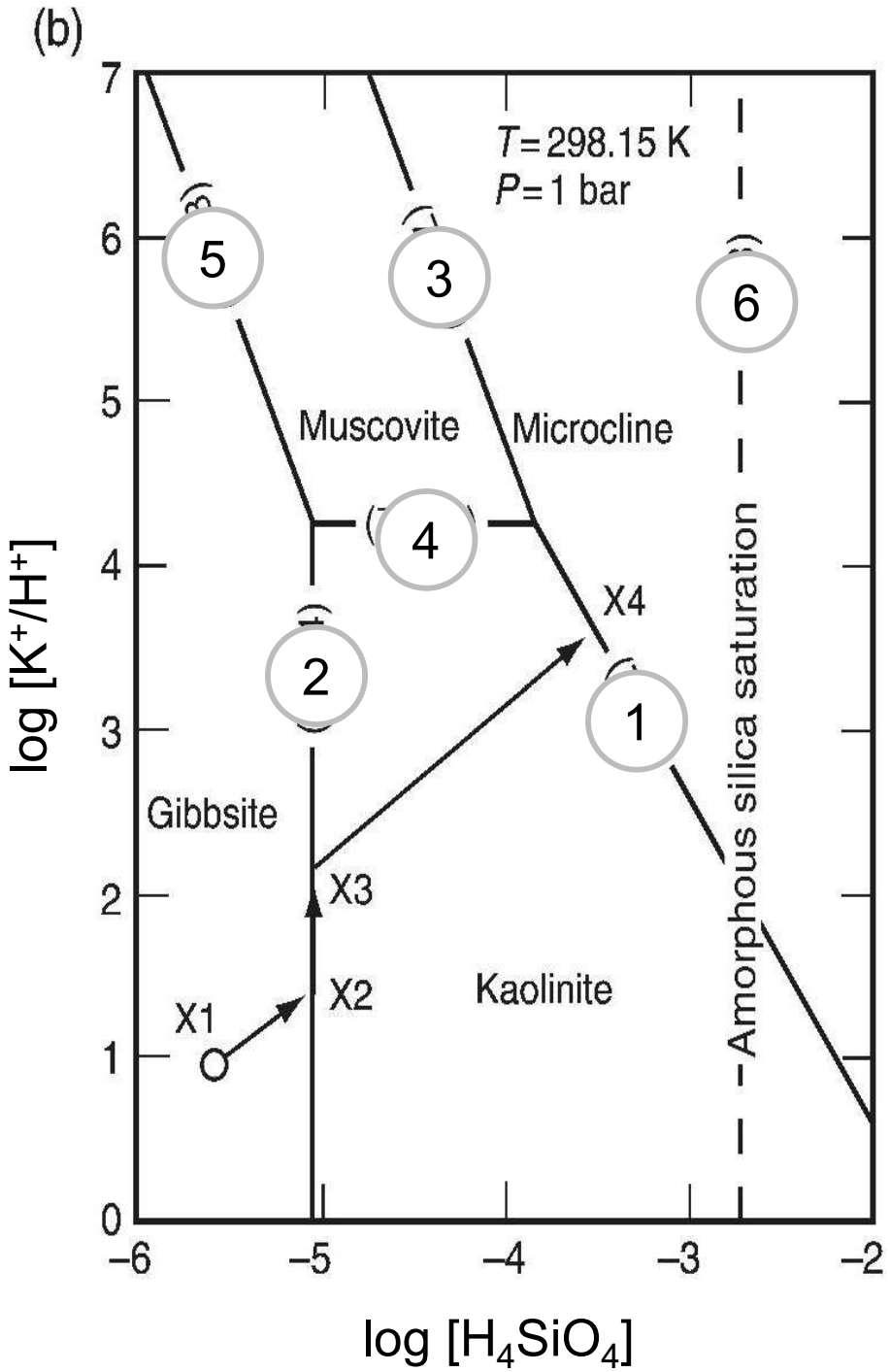
This diagram gives us the stability of each mineral and for equilibrium between pairs of minerals.

Assuming closed conditions and low water/rock ratios:

X1 fluid interacts with a rock containing microcline then gibbsite will form (note these two minerals are not stable in contact with each other). This will release K^+ and silicic acid according to:

$KAlSi_3O_8\,(s) + 7H_2O(l) + H^+(aq)$
$\leftrightharpoons\ K^+(aq) + Al(OH)_3(s) + H_4SiO_4\,(aq)$

Mineral stability diagram



At X2 microcline reacts to form kaolinite. K^+ ions and silicic acid are formed:

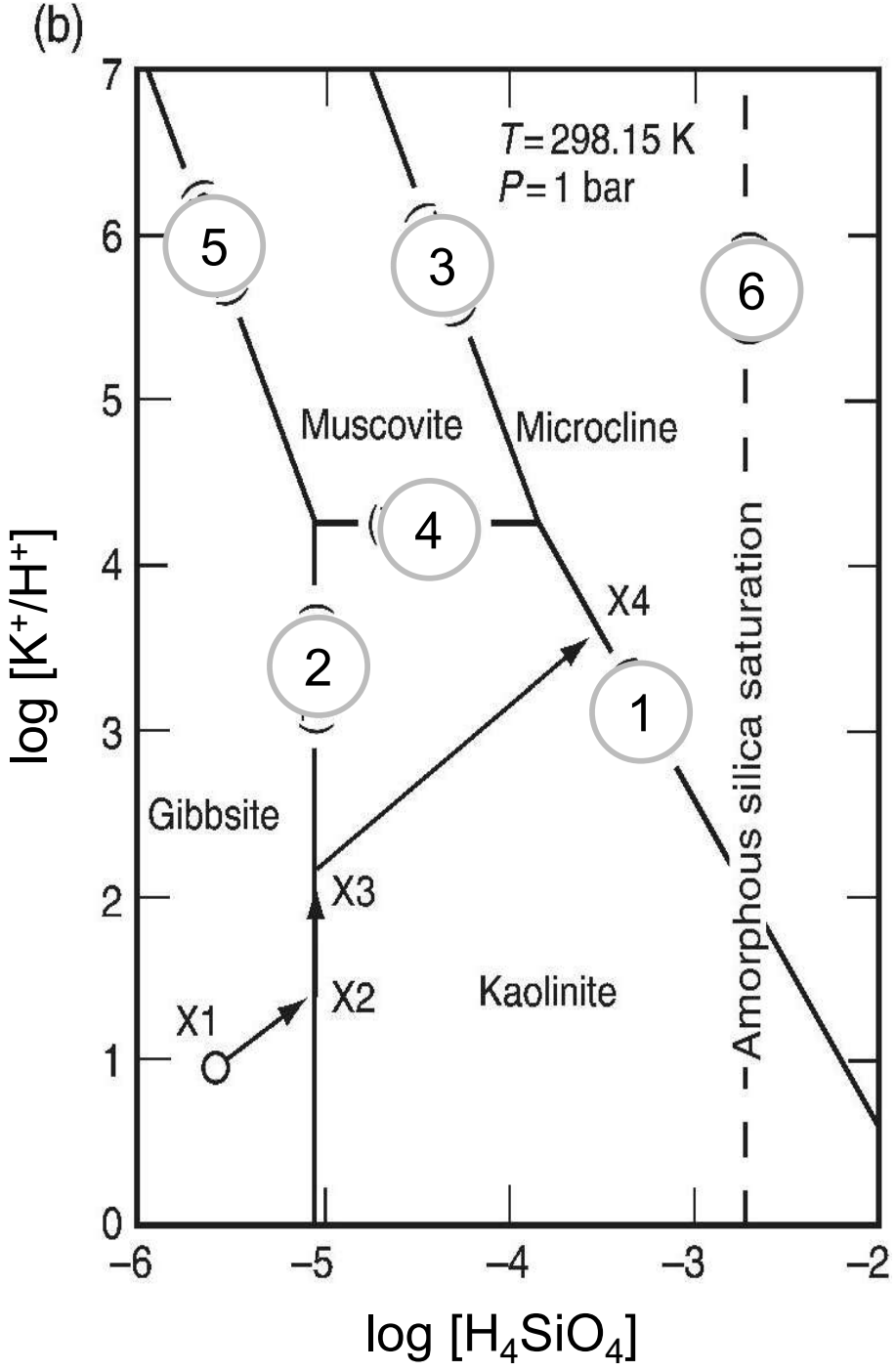
$2KAlSi_3O_8(s) + 9H_2O\ (l) + 2H^+(aq) \leftrightharpoons 2K^+(aq) + Al_2Si_2O_5(OH)_4(s) + 4H_4SiO_4(aq)$

Release of H_4SiO_4 pushes **X2** into the kaolinite stability field and gibbsite formed previously will combine with silicic acid to form kaolinite.

$2Al(OH)_3(s) + 2H_4SiO_4(aq) \leftrightharpoons Al_2Si_2O_5(OH)_4(s) + H_2O(l)$

.

Mineral stability diagram



This diagram gives us the stability of each mineral and for equilibrium between pairs of minerals.

X3 is a point when all the gibbsite is consumed then microcline reacts to form kaolinite and releases K+ and silicic acid.

At **X4** microcline and kaolinite are in equilibrium and the reaction stops.

Summary - 1

- Acidity and alkalinity are the capacity to donate or accept protons respectively.
- Alkalinity is an important measure of a natural solutions ability to buffer pH during addition of acid solution.
- A pH buffer is the capacity of a solution to “resist” changes in pH when and acid or base is added.
- The changes in pH during buffering can be approximated using the Henderson-Hasselbach equation and buffering is strongest within +/- 1 pH unit of the pK value of the acid or base.
- pH buffering in seawater is controlled mainly by the dissolution and precipitation of calcium carbonate
- In pure water the concentration of H^+ and OH^- ions are equivalent at 1×10^{-7} mol L^{-1} and the -log of this is the pH = 7.

Summary - 2

- The solubility of some important geological compounds are sensitive to changes in pH.
- Silica is much more soluble in alkaline solutions (pH>7) due to the stepwise dissociation of silicic acid.
- Precipitation and dissolution of the mineral quartz from supersaturated and undersaturated solutions respectively, is kinetically unfavourable.
- In the oceans, precipitation of silica is biologically mediated.
- Mineral stability diagrams can be used to define the environmental conditions under which each phase (mineral) is stable and what happens when a mineral is exposed to conditions outside of its stability field.

Reading – Misra: Chapter 7.5-7.7